Week_3

1.Class and object.

```
#include <iostream>

using namespace std;

class Student {

public:

    string name;

    int age;


    void display() {

        cout << "Name: " << name << endl;

        cout << "Age: " << age << endl;

    }

};


int main() {

    Student s1;   // object

    s1.name = "Ram";

    s1.age = 20;

    s1.display();
```

```
    return 0;
}
```

2.Constructor and Destructor

```
#include <iostream>
using namespace std;

class Demo {
public:
    Demo() {
        cout << "Constructor called" << endl;
    }

    ~Demo() {
        cout << "Destructor called" << endl;
    }
};

int main() {
    Demo obj;
    return 0;
}
```

3.inear search or sequential search

```
#include <iostream>

using namespace std;


int main()

{

   int n, i, key, pos = -1;


   cout << "Enter number of elements: ";

   cin >> n;


   int arr[n];


   cout << "Enter elements: ";

   for(i = 0; i < n; i++)

   {

      cin >> arr[i];

   }


   cout << "Enter element to search: ";

   cin >> key;
```

```
    // Linear Search
    for(i = 0; i < n; i++)
    {
        if(arr[i] == key)
        {
            pos = i;
            break;
        }
    }

    if(pos != -1)
        cout << "Element found at position " << pos + 1;
    else
        cout << "Element not found";

    return 0;
}
```

4.Number of ocurrences of key element

```
#include <iostream>
using namespace std;
```

```
int main()
{
    int n, i, key, count = 0;

    cout << "Enter number of elements: ";
    cin >> n;

    int arr[n];

    cout << "Enter elements: ";
    for(i = 0; i < n; i++)
    {
        cin >> arr[i];
    }

    cout << "Enter element to search: ";
    cin >> key;

    // Linear Search to count occurrences
    for(i = 0; i < n; i++)
    {
```

```
        if(arr[i] == key)
        {
            count++;
        }
    }

    if(count > 0)
        cout << "Element found " << count << " times";
    else
        cout << "Element not found";

    return 0;
}
```

5.Binary search

```
#include <iostream>
using namespace std;
int main()
{
    int n, i, key;

    cout << "Enter number of elements: ";
```

```
cin >> n;

int arr[n];

cout << "Enter sorted elements: ";
for(i = 0; i < n; i++)
{
    cin >> arr[i];
}
cout << "Enter element to search: ";
cin >> key;

int low = 0, high = n - 1, mid;
bool found = false;

while(low <= high)
{
    mid = (low + high) / 2;

    if(arr[mid] == key)
    {
        cout << "Element found at position " << mid + 1;
```

```
            found = true;
            break;
        }
        else if(arr[mid] < key)
        {
            low = mid + 1;
        }
        else
        {
            high = mid - 1;
        }
    }
    if(!found)
        cout << "Element not found";
    return 0;
}
```